



CQRS & Event Sourcing Essentials

Understanding Patterns for Scalable Systems

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Agenda



1. Introduction to CQRS
2. Event Sourcing Concepts
3. CQRS Usage Considerations
4. Summary and Takeaways

3.



Introduction to CQRS

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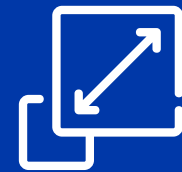


Why CQRS Exists



Conflicting Read/Write Demands

Read and write operations often require different optimizations.



Scalability Bottlenecks

Handling both workloads together can limit system scalability.



Need for Specialized Solutions

Separating models helps address unique workload challenges.



CQRS Fundamentals

Distinct Command and Query Paths

- Commands handle data changes; queries fetch information.

Independent Optimization

- Write models enforce business logic and consistency.
- Read models are tailored for fast, efficient data retrieval.

Enhanced System Efficiency

- Separation reduces bottlenecks and improves scalability.

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Read vs Write Models



Write Models Ensure Consistency

Strictly enforce business rules and data integrity.



Read Models Deliver Speed

Optimized for rapid, efficient data retrieval.



Independent Optimization

Each model tailored for its specific workload.

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CQRS Flow Diagram



Commands trigger changes in the write model

Business logic processes incoming commands



Events are generated from write operations

Events capture state changes for the system



Read model projections are updated by events

Projections enable fast, efficient queries

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Event Sourcing Concepts

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Event Sourcing Basics



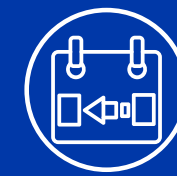
Events define system state

Every change is captured as an event



Append-only event log

Events are stored in sequence, never altered



State rebuilt by replaying events

System reconstructs current state from event history

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Benefits of Event Sourcing



Comprehensive Audit Trail

Every change is recorded as an event for full traceability.



Replay System State

Rebuild any past state by replaying the event log.



Temporal Queries Enabled

Query system history at any point in time.

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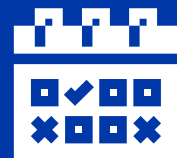


Event Store



Immutable Event Storage

Events are never changed or deleted.



Chronological Ordering

Events are stored in the order they occur.



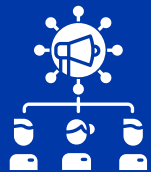
System Source of Truth

Event store maintains the definitive system record.

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Projections



Transforms Event Data

Converts raw events into structured, queryable views.



Optimized for Fast Reads

Projections enable efficient data retrieval for queries.



Supports Multiple View Models

Different projections can serve diverse business needs.

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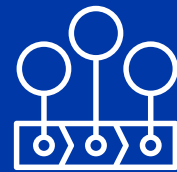


Snapshots



Boosts Recovery Speed

Snapshots allow rapid system rebuilds without full event replay.



Minimizes Performance Overhead

Reduces the need to process millions of historical events.



Streamlines System Maintenance

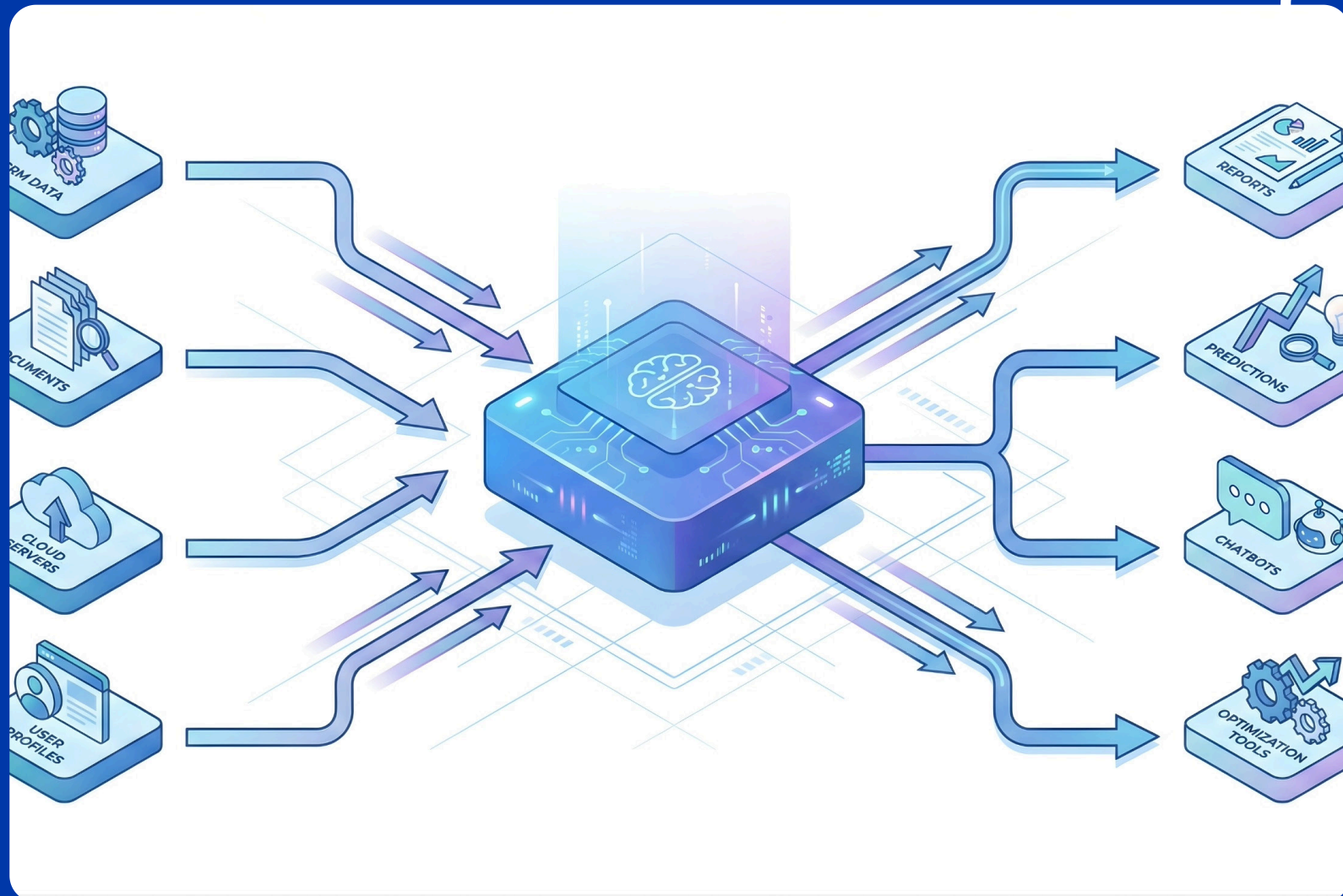
Facilitates efficient backups and restores for large event stores.

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CQRS Usage Considerations

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CQRS Without Event Sourcing

CQRS operates without event sourcing

Command and query separation does not require event logs

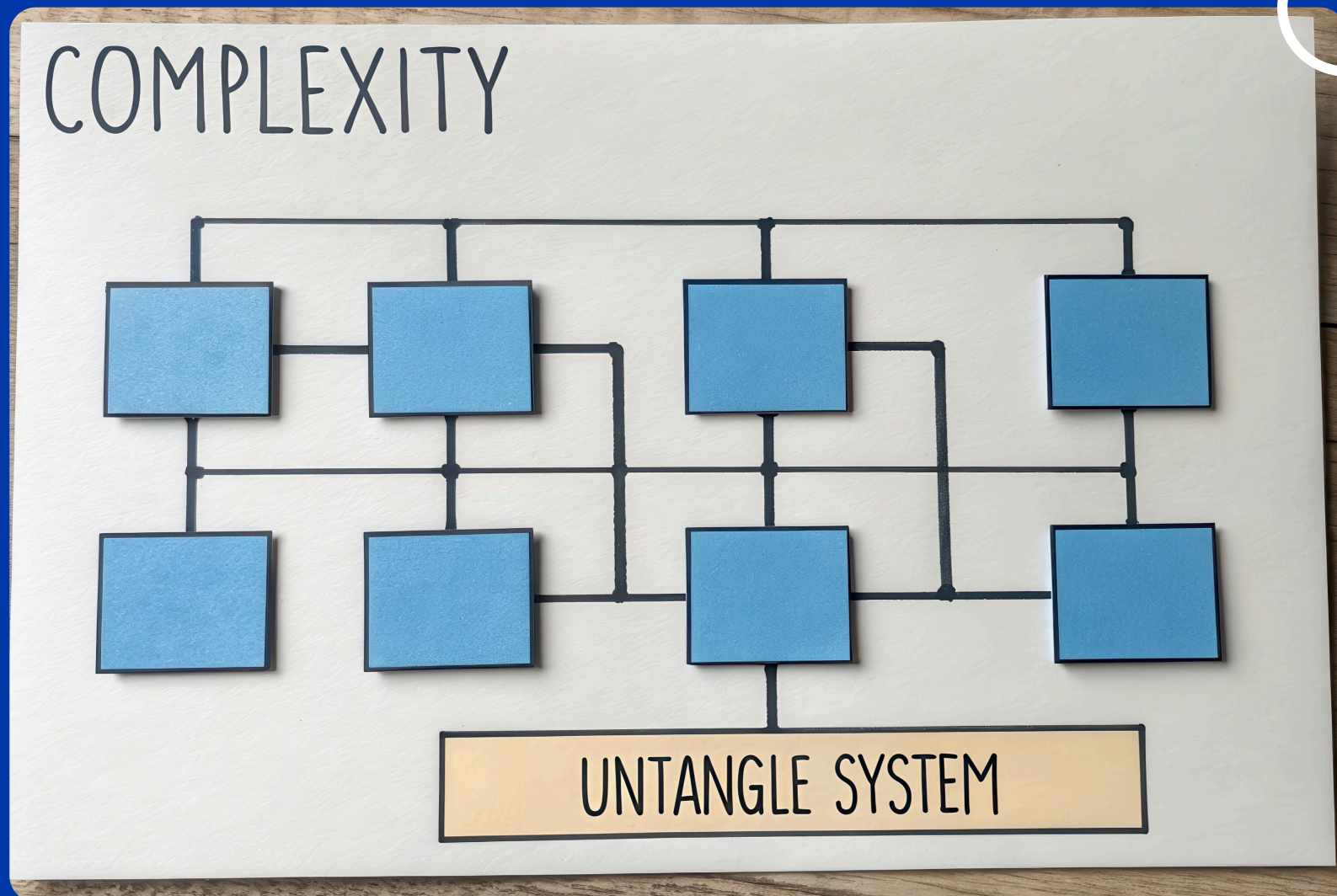
Event sourcing is an optional enhancement

Systems can choose event sourcing for added capabilities

Simpler architecture for some use cases

Avoids extra complexity when event history is not needed

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When NOT to Use CQRS

Not suited for basic CRUD systems

Adds overhead without clear benefits

Complexity can outweigh advantages

Extra layers may slow small teams

Choose simplicity when possible

Use CQRS only if system demands it

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Summary and Takeaways

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Module 09 Takeaways

CQRS drives scalability in complex systems

Separates read and write models for optimal performance

Event sourcing unlocks advanced capabilities

Enables audit trails and system history replay

Adopt only when complexity demands

Avoid unnecessary overhead for simple applications